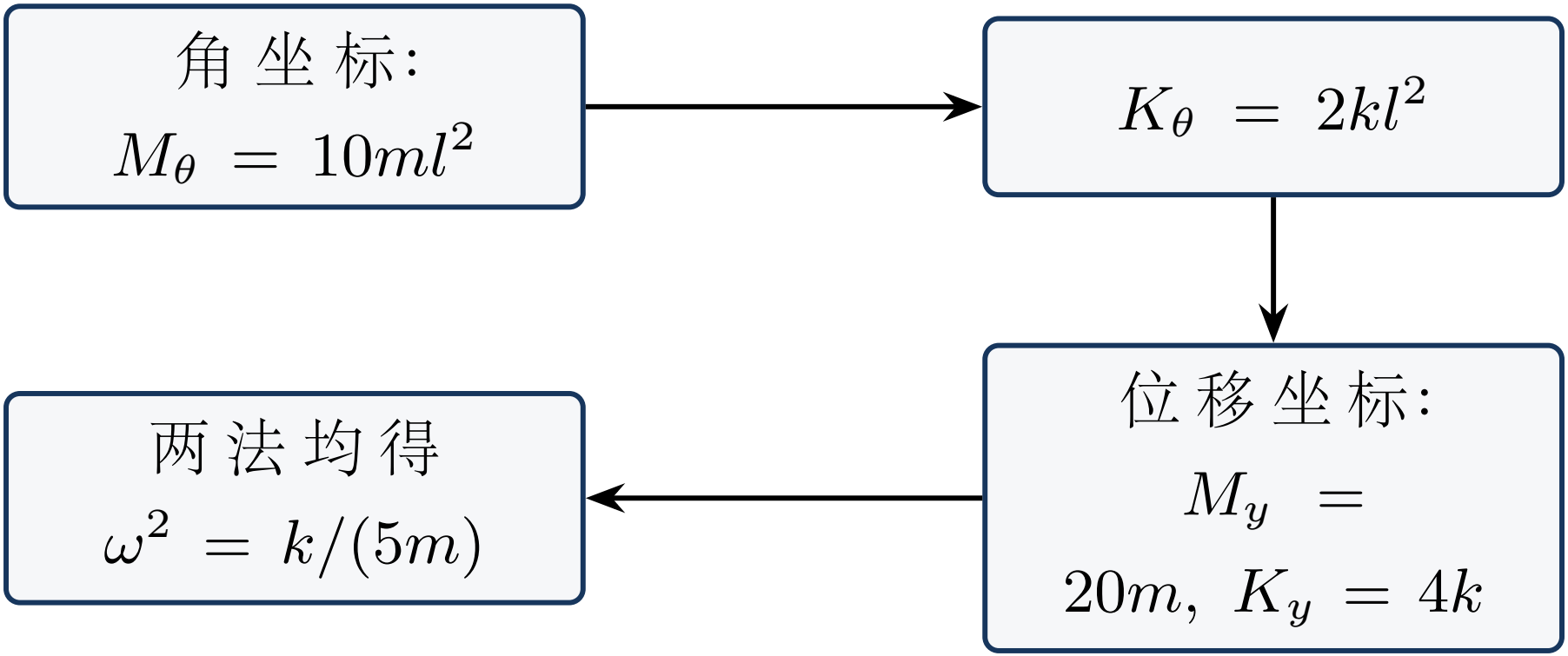

弹簧长度增量 $2y$; $U - U_0 = 2ky^2$。平衡条件给 $k = \sqrt{2}mg/l$。